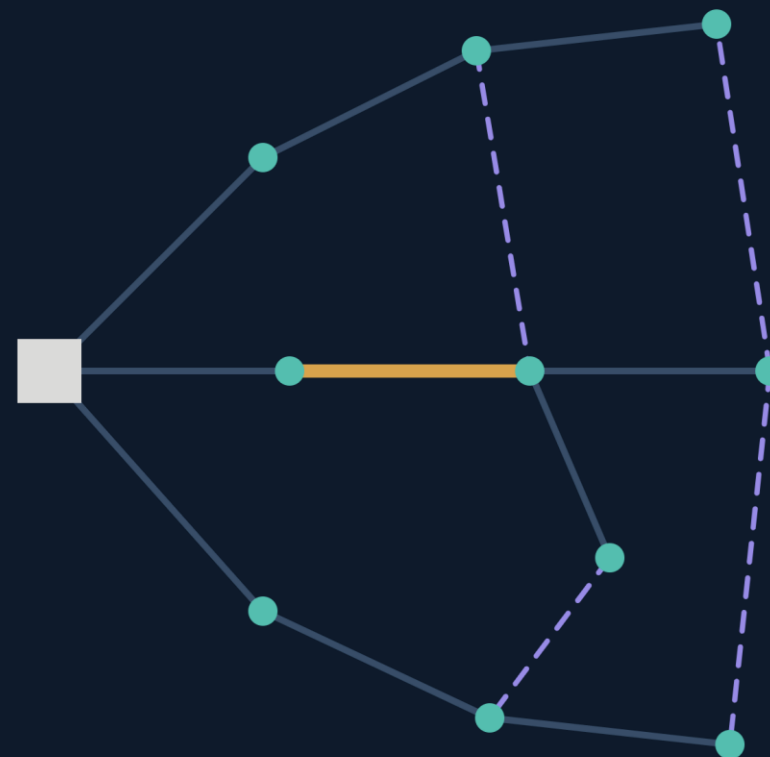


GRID EXPANSION PROPOSAL

CIQ-Expand

Congestion-Interaction QUBOs for quantum-enabled distribution grid expansion

E.ON: Europe's largest energy distribution network



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PROBLEM FRAMING

Reinforcing MV/LV grids for electrified demand

A distribution system operator must reinforce its medium- and low-voltage network so that growing load and injection stay within thermal ratings and voltage limits.

Heat pumps

EV charging

Rooftop PV

We choose a subset of candidate lines K to add: new feeder ties, parallel cables, or substation-to-load-centre links.

THE DECISION

$$\mathbf{x} \in \{0, 1\}^{|K|}$$

$$\min \sum c_k x_k + \rho \cdot Q(\mathbf{x})$$

Investment cost plus residual congestion $Q(\mathbf{x})$: the overload left after flows redistribute, summed over all lines and weighted operating scenarios.

REPORTED FOR EACH PLAN

Lines added · objective value · congestion events (loading above 100%) · near-congestion events (above 90%)

Three sources of difficulty compound



Interacting decisions

A line's value depends on which other lines are built. Two ties relieving one feeder are substitutes; lines that help only by closing a loop together are complements.



Non-local physics

Kirchhoff's laws redistribute flow grid-wide, so a new line can overload another (Braess's paradox). Exact MILPs need big-M constraints with weak LP relaxations.

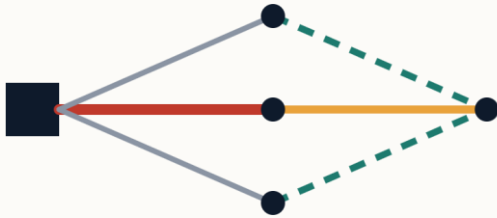


Scenario and topology coupling

One plan must work across many operating snapshots. Radial operation adds Steiner-tree and reconfiguration structure, which is strongly NP-hard.

Result today: planners evaluate few options and settle for conservative, costlier plans.

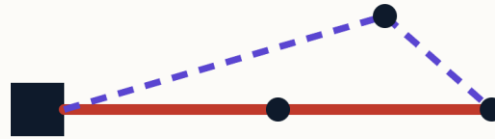
The sign of each coefficient tells the story



$$J(k, m) > 0$$

Substitutes

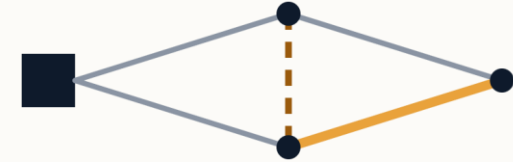
Either tie relieves the overload, so building both is redundant.



$$J(k, m) < 0$$

Complements

Each line is useful only when the other closes the loop.



$$h(k) > 0$$

Braess-type line

Adding k pushes flow onto another line and raises its loading.

A mix of all three produces frustrated energy landscapes that trap classical local search.

KEY OBSERVATION

Residual congestion is dominated by low-order interactions between electrically close candidates.

So the optimisation core is a sparse, frustrated QUBO whose locality is inherited from the grid topology.

LOCALITY

Short-range couplings map onto native qubit connectivity with shallow circuits.

FRUSTRATION

Mixed-sign couplings create rugged landscapes where local search gets trapped.

What we will and will not claim

Advantage is tested, not assumed

- Certify classical hardness first, following the QOBLIB methodology
- Compare against the best classical solvers at equal wall-clock time
- Report negative results as well as positive ones

Motivated by evidence of QAOA scaling advantage on a classically hard problem and utility-scale circuits on 100+ qubits.

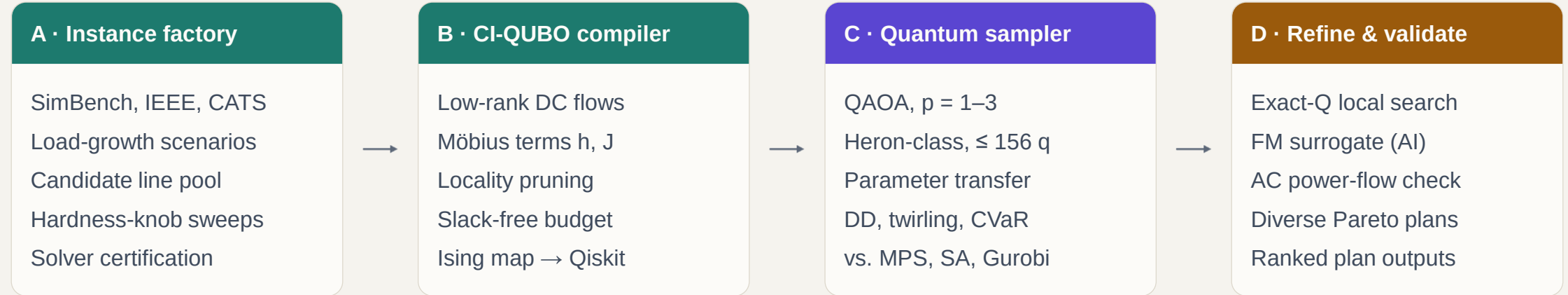
Value even without speedup

- A sampler returns a portfolio of diverse near-optimal plans
- Multi-objective sampling yields a cost–congestion Pareto front
- Engineers choose, because permitting and constructability decide what gets built

A useful planning tool regardless of the quantum outcome.

TECHNICAL APPROACH

A four-stage hybrid pipeline



Feedback loop, D back to B: couplings learned from exact evaluations update the QUBO for 2–5 active-learning rounds.

■ Classical HPC ■ Quantum processor ■ AI-driven refinement

A certified hard-instance factory

Grids

SimBench MV/LV (German grids where E.ON operates DSOs), CIGRE, IEEE 33/69-bus, CATS and TAMU, plus synthetic look-alikes.

Scenarios

Heat-pump, EV and PV growth multipliers, clustered to 4–24 snapshots.

Two formulations

Exact multi-scenario MILP (F1) and the quantum-native CI-QUBO (F2).

Certification

Gurobi, CPLEX, HiGHS/SCIP at 1 h and 4 h, 5 seeds. Hard means not proven optimal in 1 h, or exponential runtime growth. Exported in QOBLIB format with best-known solutions.

Hardness knobs swept

Knob	Range
Load growth α	1.0–2.0, fine near congestion onset
Candidate density $ K / L $	0.2–1.5
Loop-closing share	0–60% of candidates
Budget tightness $B/\Sigma c$	0.05–0.40
Scenario count $ S $	4–24 snapshots
Cost heterogeneity	$\pm 50\%$ noise on length cost

The Congestion-Interaction QUBO

$$\hat{Q}(x) = Q(0) + \sum h_k x_k + \sum J_{km} x_k x_m$$

$$h_k = Q(e_k) - Q(0) \quad J_{km} = Q(e_k + e_m) - Q(e_k) - Q(e_m) + Q(0)$$

Adding a line is a rank-one susceptance update, so flows follow exactly from base PTDFs via Sherman–Morrison–Woodbury, with no refactorisation.

Exact where it matters

$\hat{Q}$ equals Q for every plan with at most two additions.

Cheap to build

11,326 low-rank evaluations per scenario for $n = 150$, embarrassingly parallel.

Interpretable

$J > 0$ substitutes, $J < 0$ complements, $h > 0$ Braess-type lines.

Sparse

Couplings decay with electrical distance; pruning small $|J|$ leaves a sparse graph.

Cost Hamiltonian and budget encodings

$$H(x) = \sum c_k x_k + \rho \cdot Q^{\wedge}(x) + P_B(x)$$

DENSE TIERS (S, M)

Unbalanced penalisation

$P_B = -\lambda_1 g + \lambda_2 g^2$, with $g = B - \sum c_k x_k$. No slack qubits are needed, so one qubit equals one candidate line.

HARDWARE-NATIVE TIER (L)

Lagrangian budget term

$\lambda \sum c_k x_k$ keeps the coupling graph sparse. Sweeping λ traces the cost-congestion Pareto front, and sample repair guarantees feasibility.

Ising map $x_k = (1 - Z_k)/2$. Cardinality budgets (at most m lines) can be enforced exactly with XY mixers. Interactions among three or more lines are handled in Stage D.

QAOA sampling on Heron-class hardware

Algorithm. QAOA at depth $p = 1\text{--}3$ via Qiskit Runtime SamplerV2 on heavy-hex processors, up to 156 qubits.

Hardware-aware compilation. For the largest tier only SWAP-free couplings are kept, giving CZ depth 6 per layer; dropped weak couplings return in classical scoring.

Parameter transfer. Angles optimised by statevector on ≤ 30 qubits and transferred to large instances; short CVaR fine-tuning only where needed.

Error suppression. Dynamical decoupling, Pauli twirling and readout mitigation. We need good samples, not precise expectation values.

Execution tiers

Tier	Qubits	p	CZ / layer	Shots	Purpose
S	16–32	1–3	150–600	10^4	Exact optimum known
M	48–100	1–2	250–700	$2 \cdot 10^4$	Scaling curves, ablations
L	127–156	1–2	≤ 352	10^5	Hardness regime, MPS crossover

QPU budget $\approx 2 \text{ h (S)} + 3 \text{ h (M)} + 6\text{--}8 \text{ h (L)} + 20\% \text{ contingency} \approx 12\text{--}15 \text{ QPU-hours}$.

STAGE D · AI-DRIVEN REFINEMENT

Refine, learn, and validate every plan

STEP 1

Refine

Every sample is repaired for budget feasibility, polished by steepest-descent bit-flip search on the exact $Q(x)$, and scored.



STEP 2

Learn

Elite plans train a factorization-machine surrogate. It has QUBO form, so its coefficients update the CI-QUBO and capture higher-order interactions.



STEP 3

Validate

AC power flow in pandapower across all scenarios: thermal loading, EN 50160 voltage band 0.9–1.1 p.u., losses, and N-1 for MV.

OUTPUT

Lines added, cost, congestion and near-congestion reduction, and a ranked, diverse Pareto set of plans.

FEASIBILITY

Resources for a 16-week proof of concept

10–15

QPU-hours on IBM Heron-class,
about 250 sampler jobs

≤ 156

qubits on heavy-hex, Qiskit-native
code

≤ 4 h

end to end per instance, excluding
queue

16 wk

PoC, with QPU access in weeks 9–14

Classical compute and software

1 CPU node (64 cores, 256 GB) plus 1–2 datacentre GPUs. Python, Qiskit 2.x, Aer, pandapower, SimBench, PyTorch; commercial MILP solvers serve only as baselines.

Circuit budget, largest tier

About 176 couplers give ≤ 352 CZ per layer at CZ depth 6, about 700 at $p = 2$. Fidelity will be low but not negligible; we rely on sample quality after local search, and ablation M4 measures what survives.

RISKS

Key risks and how we contain them

Risk	L / I	Mitigation
Noise erases the quantum signal at 100+ qubits	H / M	Hardware-native sparsification, $p \leq 2$, DD and twirling; ablation M4 measures it; fall back to tier M
Second-order surrogate misses higher-order interactions	M / H	Exact re-scoring of every sample, FM active learning, optional 3-local terms; go/no-go gate M2
Instances not hard for Gurobi at NISQ sizes	M / H	Systematic knob sweeps, F1 hardness, QOBLIB-style hard-for-size reporting
Shallow circuits are classically simulable	M / M	MPS and light-cone study (M5); move to $p \geq 2$; claim only what survives
DC approximation inaccurate on LV feeders	M / M	LinDistFlow sensitivities; AC validation of all final plans
QPU queue delays or late E.ON data	L / M	Simulator first, parameter transfer, backend-agnostic code; open SimBench grids meanwhile

L / I = likelihood / impact

EXPECTED IMPACT

From a handful of options to thousands of scored plans

Planning aspect	Current practice	With CIQ-Expand
Option generation	Rules and experience; a few candidates iterated through simulation	Systematic candidate pools; thousands of plans physics-scored automatically
Line interactions	Assessed implicitly, one option at a time	Explicit h/J sensitivities flag substitutes, complements and Braess-type lines
Scenario coverage	Limited by spreadsheets and engineer effort	All scenarios scored jointly for every candidate plan
Planner output	One recommended plan	Ranked, diverse Pareto set with cost, congestion, losses and AC checks
Quantum evidence	Anecdotal or vendor claims	Certified-hard instances, ablations and MPS crossover, open and reproducible

Outputs in pandapower JSON, MATPOWER and CSV import into existing grid-analysis tools, keeping the engineer in the loop.

What success looks like

$\geq 90\%$

fewer congestion events than the
unreinforced grid

$\leq 5\%$

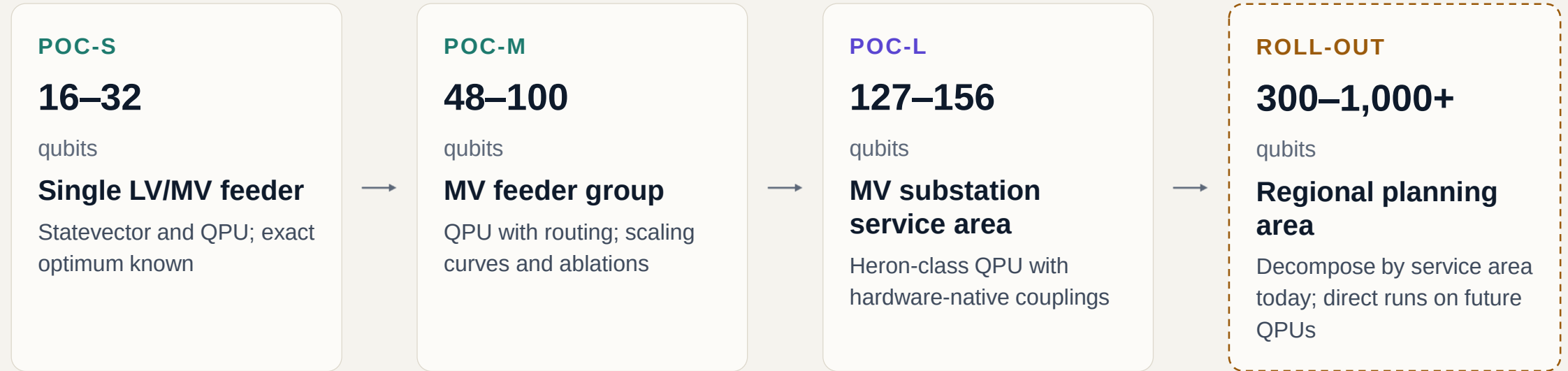
investment cost gap to the best
classical plan

≥ 10

distinct plans within 2% of the best-
known objective

Plus the first open, certified-hard grid-expansion library in QOBLIB format, benchmarked against Gurobi, SA, tabu and MPS whether the result is positive or negative.

Qubit count grows linearly with the candidate pool



The pipeline carries over unchanged to larger QPUs. We aim for a joint scientific publication with E.ON.

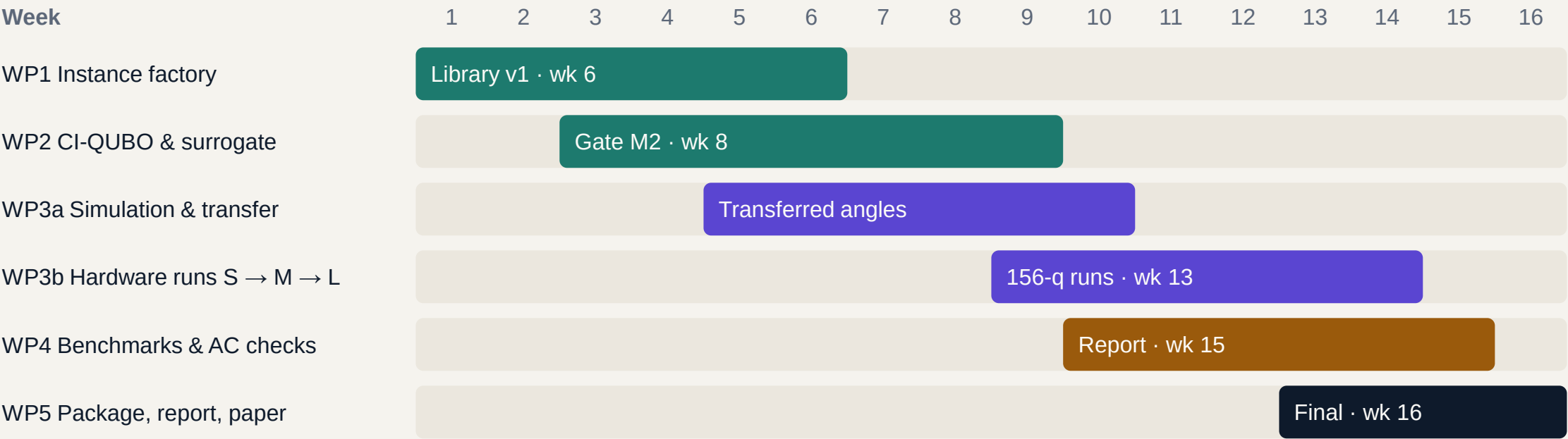
VALIDATION PLAN

Seven metrics, reported whatever the outcome

ID	Metric	Success criterion
M1	Instance hardness	≥ 10 certified-hard instances in ≥ 3 size classes, ≥ 3 with 100–156 variables
M2	Surrogate fidelity (gate, wk 8)	Spearman $\rho \geq 0.9$ and top-10 overlap $\geq 80\%$; otherwise add third-order terms
M3	Solution quality	Gap $\leq 1\%$ for $n \leq 32$ and $\leq 3\%$ for $n \geq 100$; stretch: match Gurobi at equal time
M4	Quantum contribution	Significant gain vs. random, SA and tabu samples ($p < 0.05$), reported either way
M5	Classical simulability	Locate sizes and depths where MPS at equal runtime loses to hardware
M6	Grid outcome	$\geq 90\%$ fewer congestion events, no new voltage violations, losses reported
M7	Engineering	≤ 4 h per instance excluding queue; one-command Docker reproduction

PHASE 2 WORK PLAN

Sixteen weeks, one go/no-go gate



Fallback if gate M2 fails: Benders decomposition of F1 with a QUBO master problem.

DELIVERABLES

Open, reproducible, and re-executable by E.ON

- D1** Open hard-instance library with best-known solutions, in QOBLIB format
- D2** Qiskit-native code, Docker image and an E.ON re-execution guide
- D3** Benchmark report covering M1–M7, including negative results
- D4** Plan reports for E.ON anonymised subgrids: lines added, cost, congestion before and after, AC checks
- D5** Joint paper draft

ONE-COMMAND REPRODUCTION

```
ciq run --instance <id> --backend <ibm_backend | aer_mps>
```

Every quantum job is stored as a Qiskit QuantumCircuit in QPY together with its transpilation settings, inside the provided Docker image.

```
ciq.instances · ciq.qubo · ciq.solvers ·  
ciq.validate · ciq.bench
```

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